

ECON-1020 Introductory Macroeconomics

Chapter 9 Unemployment and Its Natural Rate

Shutao Cao

2025-09-22

Trent University

<http://www.trentu.ca>

Unemployment and Employment

Natural rate of unemployment

Taking Stock

Appendix: Employment Effects of AI

Unemployment and Employment

Natural rate of unemployment

Taking Stock

Appendix: Employment Effects of AI

Current week:

- Textbook Chapter 9

Next week:

- Textbook Chapter 7

Unemployment and employment is an important issue in macroeconomics

- Labor income accounts for about 60% of aggregate income (GDP)
 - The rest of GDP is accounted for by capital income and profits
- Supplying labor is the primary source of income for the majority
- Macroeconomic policies often directly impact the labor market

Unemployment rate, GDP growth, and inflation rate are the most important macroeconomic indicators tracked by policy makers and investors

Unemployment and Employment

Definitions

Employment: The number of people who have a paid job, or work in own businesses

Unemployment: The number of people who do not have a job but want one, including those who

- have looked for work in the past four weeks, or
- are temporarily laid-off and are available for work, or
- do not have a job but expect to start a job within four weeks from a reference period

The **labor force**: the total number of workers

- It consists of the employed and the unemployed

A person who is neither employed nor unemployed is **not in the labor force**

The **working-age population** includes all persons aged 15 years and over residing in Canada, excluding members of full-time regular army forces and institutionalized persons (inmates, patients in hospitals or nursing homes)

Labor market indicators

Labor force = employment + unemployment

Unemployment rate = $\frac{\text{Number of unemployed}}{\text{Labor force}} \times 100\%$

Labor force participation rate = $\frac{\text{Labor force}}{\text{Working-age population}} \times 100\%$

Employment rate = $\frac{\text{Number of employed}}{\text{Working-age population}} \times 100\%$

Some call the employment rate the employment-population ratio

Other labor market indicators:

- under-employment rate
- involuntary part-time rate

Is the unemployment rate a good measure?

The unemployment rate is widely used as a measure of the **labor market tightness**—the degree of difficulty firms face in hiring workers

Unemployment rate can mis-measure the labor market tightness:

- First, **discouraged workers** give up and stop searching for work, they are not counted in the labor force
 - If there are more discouraged workers, unemployment rate drops
 - For example, if unemployment rate is $\frac{10}{100} = 0.1$. Now 1/10 of the unemployed stop searching for work. The unemployment rate becomes $\frac{10-1}{100-1} = 0.091$

- Second, unemployment rate tells little how intensively the unemployed search for work
 - At a high unemployment rate, an unemployed worker may not intensively search for work, making it harder (not easier) to find a worker for firms hiring

A look at the labor market in Canada

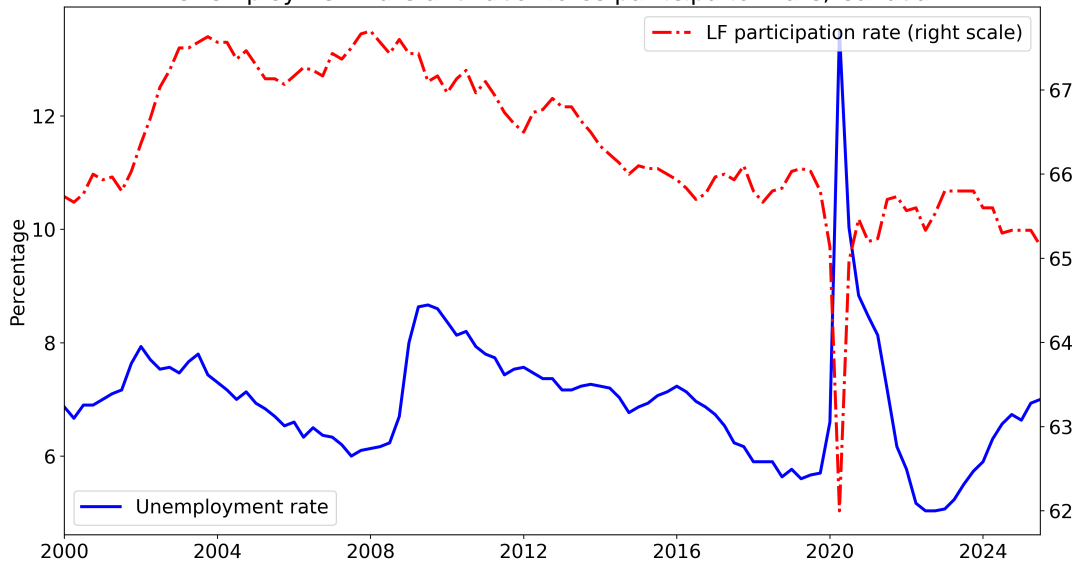
Data source:

- **Labor Force Survey** (LFS), monthly, sampling 56,000 households
- The LFS asks working-age people about their employment status
- **Labor Market Indicators** at Statistics Canada

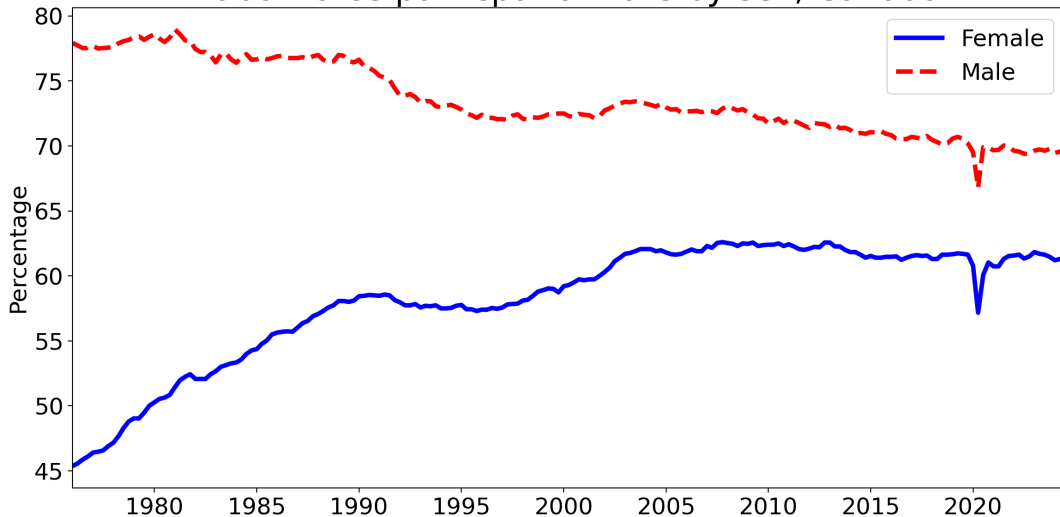
Labor market snapshot: August 2025 (Statistics Canada Table 14-10-0287)

- Working-age population: 34.64 million
- Labor force: 22.55 million, of which:
 - Employment 20.95 million; Unemployment 1.60 million
- Unemployment rate, 7.1%
- Labor force participation rate, 65.1%
- Employment rate, 60.5%

Unemployment rate and labor force participation rate, Canada



Labor force participation rate by sex, Canada



Data source: Statistics Canada

Natural rate of unemployment

Types of unemployment

There are always some unemployment rate

- over 1976-2025, the lowest unemployment rate was 4.8% (July 2022)

We define three types of unemployment, each associated with a cause:

- Frictional unemployment
- Structural unemployment
- Cyclical unemployment

We then define the natural rate of unemployment

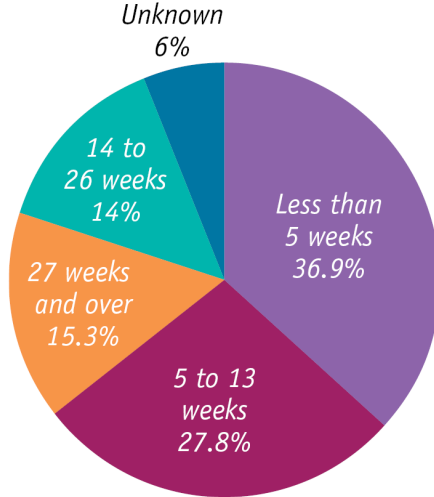
Frictional Unemployment

It takes time for workers to search for jobs best suited for them

Frictional unemployment refers to the unemployment that results from the process of searching and matching workers and jobs

Some frictional unemployment is inevitable

Time since being unemployed, January 2020, Canada



Krugman et al., Macroeconomics, Canadian 4e, © 2021 Worth Publishers

Frictional unemployment in a constantly changing economy

Over the past century, advanced economies observed sectoral shifts

- Employment in agriculture observed declines, followed by the employment declines in manufacturing
- Meanwhile, employment in services has been rising
- Globalization also shifted production across countries

Sectoral shifts cause frictional unemployment, e.g.,

- the rust belt in the U.S. midwest
- high unemployment in Peterborough in the 2000s due to declines in manufacturing plants

The current wave of adopting artificial intelligence will cause frictional employment

Acemoglu et al 2012, Journal of Labor Economics:

Even before the Great Recession, US employment growth was unimpressive. Between 2000 and 2007, the economy gave back the considerable employment gains achieved during the 1990s, with a historic contraction in manufacturing employment being a prime contributor to the slump. We estimate that import competition from China, which surged after 2000, was a major force behind both recent reductions in US manufacturing employment and—through input-output linkages and other general equilibrium channels—weak overall US job growth. Our central estimates suggest job losses from rising Chinese import competition over 1999–2011 in the range of 2.0–2.4 million.

Employment insurance (EI) is a government program that partially protects workers' incomes when they become unemployed

More generous EI benefits tends to raise the unemployment rate

Structural unemployment

Structural unemployment results from that the number of jobs available in the labor market is insufficient to provide a job for everyone wanting one

It occurs when the quantity of labor supplied exceeds the quantity demanded for labor

This often helps explain longer duration of unemployment

- E.g., many jobs disappear following the closure of a big firm

Some causes of structural unemployment

- Minimum wages
- Labor unions
- Efficiency wages

Minimum wages

Recently, Ontario government raised the minimum wage rate

Effect on the labor market?

- Theoretically, raising minimum wages tends to raise unemployment
- But economists disagree among themselves on such effect

Unions

Does increased union membership raise unemployment?

- Possible. Unions are barriers to hiring and firing workers

In 2022, union membership rate (share of workers who are members of unions):

- 29% to 30% in Canada
- 10% in the United States

Unemployment rate (August 2025): Canada 7.1%, the U.S. 4.3%

The efficiency-wage theory

Efficiency wages: wages that employers set above the equilibrium wage in order to increase worker productivity (performance)

Firms pay high wages to retain workers with good quality, and/or out of concerns of the worker morale

The unemployment may arise from efficiency wages

- Evidence? Firms tend to prefer to lay off workers instead of cutting salaries

Natural rate of unemployment

The **natural rate of unemployment** is the rate of unemployment that the economy normally experiences

The natural rate of unemployment is often called the **non-accelerating inflation rate of unemployment** (NAIRU), because it is the rate of unemployment that does not put any pressure on inflation

The natural rate of unemployment mostly corresponds to frictional unemployment and structural unemployment

- Factors affecting natural rate of unemployment are those affecting frictional unemployment and structural unemployment

The natural rate of unemployment is **unobserved**, must be estimated from the data

Central banks regularly estimate the natural rate of unemployment

- Example: In 2016 to 2018, Federal Reserve kept adjusting (lowering) its estimates of NAIRU

In Canada, the natural rate of unemployment about 6% in recent decades

Cyclical unemployment

Cyclical unemployment refers to the deviation of actual unemployment rate from the *natural rate*

- It is closely associated with short-run downturns and upturns of the economy

The cyclical unemployment rate = actual unemployment rate - natural rate of unemployment.

Full Employment

- **Full employment** is a situation in which the (actual) unemployment rate equals the natural rate of unemployment
- Thus, with full employment, the economy still has unemployment

Taking Stock

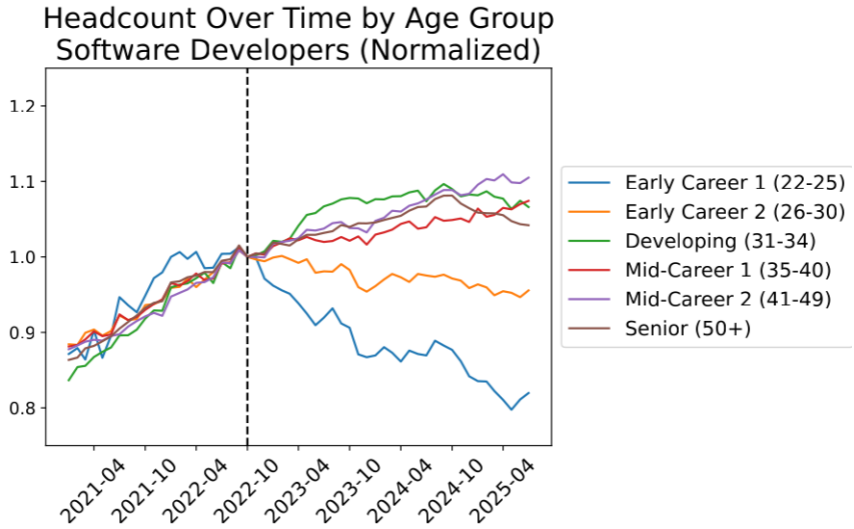
- Unemployment rate
- Types (causes) of unemployment
 - Frictional
 - Structural
 - Cyclical
- Effects of unions, EI and minimum wage
- Natural rate of unemployment

Appendix: Employment Effects of AI

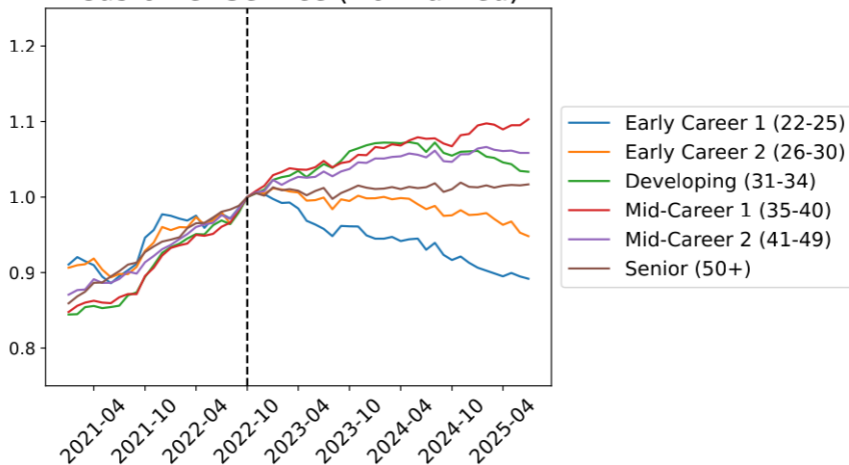
Brynjolfsson et al (2025)

- Use data from ADP, the largest payroll processing firm in America
- Present six facts of effects of AI on employment

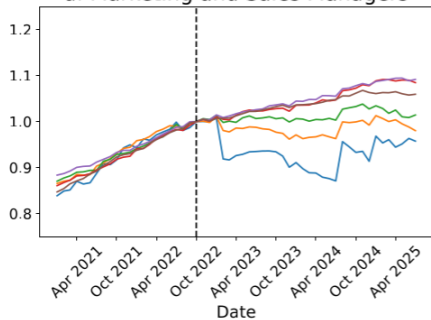
Fact 1: Employment for young workers has declined in AI-exposed occupations



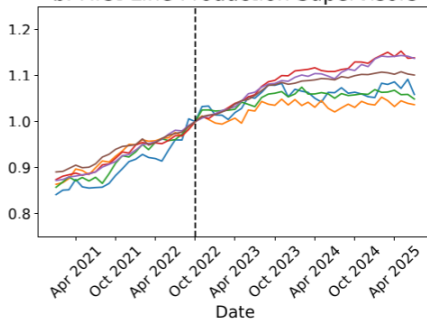
Headcount Over Time by Age Group Customer Service (Normalized)



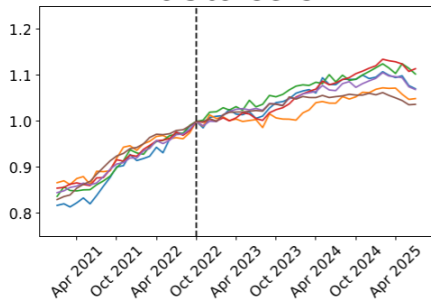
a. Marketing and Sales Managers



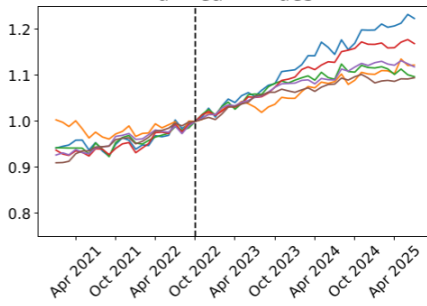
b. First-Line Production Supervisors

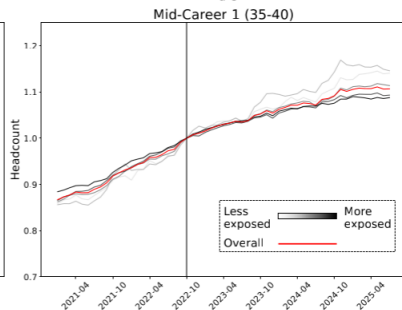
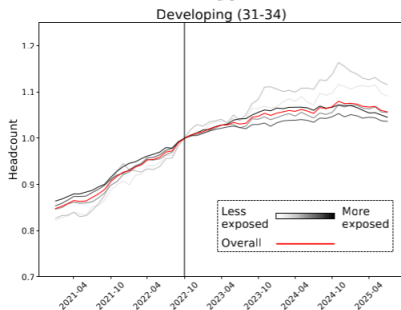
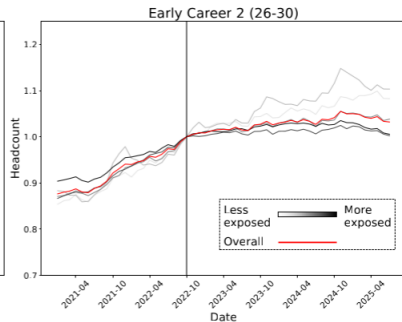
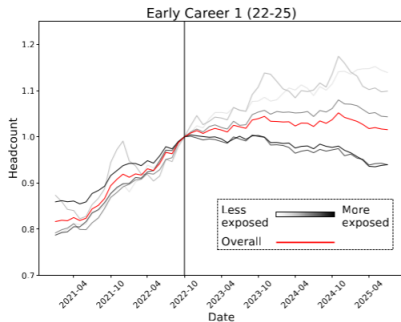


c. Stock Clerks



d. Health Aides





Other facts

Fact 2: Overall employment continues to grow, employment growth for young workers has been stagnant

Fact 3: Entry-level employment has declined in applications of AI that automate work, with muted changes for augmentation

Fact 4: Employment declines for young, AI-exposed workers remain after conditioning on firm-time shocks

Fact 5: Labor market adjustments are visible in employment more than compensation

Fact 6: Findings are largely consistent under alternative sample constructions